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July 2025

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Anterior cruciate ligament tears among football players. Current Reviews in Musculoskeletal Medicine, 18(5): 183-189, 2025.

Assessing the validity of wearable inertial sensors in evaluating joint kinetics and hamstring musculotendon mechanics at various running speeds. Medicine and Science in Sports & Exercise, Ahead of Print: 1-50, 2025.

Associations of collegiate football career and incident concussion with players' health: A longitudinal study from the CARE Consortium. Sports Medicine, Ahead of Print: 1-10, 2025.

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College-to-NFL stadium turf transitions as a risk factor for lower extremity non-contact injuries in rookie players: A 13-year cohort analysis. Healthcare, 13:1415, 2025.

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Data everywhere, insight nowhere: A practical quadrant-based model for monitoring training load vs response in elite football. Sport Performance Science Reports, May: 1-14, 2025.

Dietary carbohydrates influence the performance outcomes of short-term heat acclimation. Medicine and Science in Sports & Exercise, Ahead of Print: 1-40, 2025.

Dilemma in the treatment of sports injuries in athletes: Tendon overuse, muscle strain, and tendon rupture. Scandinavian Journal of Medicine & Science in Sports, 35:e70026, 2025.

Effect of newly required American football headgear on thermoregulation during exercise in the heat. Sports Engineering, 28:19, 2025.

Effects of different velocity loss thresholds in full squat with and without blood flow restriction on strength gains, neuromuscular adaptations, and muscle hypertrophy. Scandinavian Journal of Medicine & Science in Sports, 35:e70090, 2025.

Effects of position and injury status on associations between preseason workload and heart-rate variability profiles in American college football players. International Journal of Sports Physiology and Performance, Ahead of Print: 1-8, 2025.

Focused review of sleep wearable technology and its application in sports performance. Current Sports Medicine Reports, 24(6): 171-176, 2025.

From the Field to the Screen: A scoping review of video analysis as a tool for understanding thigh muscle-tendon injuries in football. Clinical Journal of Sport Medicine, Ahead of Print: 1-12, 2025.

Improving sleep quality to enhance athletic activity – The role of nutrition and supplementation: A mini-short review. Nutrients, 17:1779, 2025.

Irregular sleep/wake patterns in student-athletes exposed to early morning training. Journal of Sports Sciences, Ahead of Print: 1-11, 2025.

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Kinematics and kinetics of pick-up acceleration. International Journal of Sports Science & Coaching, Ahead of Print: 1-9, 2025.

Maximum-speed single-leg bridge test for concentric functional assessment and exercise in an athlete with recurrent hamstring strain injuries: A case report. International Journal of Sports Physical Therapy, 20(5): 716-726, 2025.

Monitoring dietary supplements for undeclared diuretics as a strategy for preventing unintentional doping. Processes, 13:79, 2025.

No association between fencing response after concussion and recovery time, concussion severity, or clinical outcomes in professional American football players. Neurosurgery, Ahead of Print: 1-8, 2025.

Return to performance: Machine learning insights into how absence time following muscle injuries affects match running performance in LaLiga soccer players. Biology of Sport, 42(4): 275-286, 2025.

Squatting biomechanics after femoroacetabular impingement surgery: A systematic review and meta-analysis. Clinical Journal of Sport Medicine, Ahead of Print: 1-12, 2025.

The effect of heat acclimation on critical environmental limits and rate of rectal temperature change. Journal of Applied Physiology, 138: 1150-1160, 2025.

The relationship of infrared thermography with the identification of changes in biological markers after exercise: A systematic review. Journal of Bodywork and Movement Therapies, Ahead of Print: 1-28, 2025.

Training alters joint power distributions during running in National Football League draft preparation players. Sports Biomechanics, Ahead of Print: 1-19, 2025.

Under pressure: Integrating machine learning to quantify clutch moments in American football. International Journal of Sports Science & Coaching, Ahead of Print: 1-9, 2025.

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A causal and replication analysis of claims that jet lag affects team sport performance. American Journal of Epidemiology, Ahead of Print: 1-29, 2025.

Abstract - It is broadly accepted that jet lag impacts human performance in sport; however, this has never been validated within a causal inference framework. Here, we used the potential outcomes framework to determine if jet lag, conditional on game time, causes collegiate football teams to under- or over-perform expectations. We also attempted to replicate seminal non-causal analyses in professional football. Nine collegiate football seasons (2013-2022, omitting 2020), for a total of 6,245 games were analyzed. A generalized additive model with penalized splines and independence weights was used to evaluate interactive causal effects of hours gained/lost in travel and game time on the probability of away teams beating the Las Vegas spread. Furthermore, we used our collegiate cohort to exactly replicate two previous studies purporting to show that NFL teams are adversely affected by jet lag. Jet lag's effects were compatible with the null hypothesis in both the causal analysis (P=0.133) and non-causal replications. Practically, if the effects of jet lag are unclear, then it is also unclear whether interventions to treat jet lag in elite sport are warranted for teams crossing fewer than three time zones if the primary goal is to optimize winning.

Anterior cruciate ligament tears among football players. Current Reviews in Musculoskeletal Medicine, 18(5): 183-189, 2025.

Purpose of review: Anterior Cruciate Ligament (ACL) tears are one of the most common causes of lost playing time in American football athletes. Recently, there has been a push to get athletes back to sport faster. As such, numerous studies have evaluated management and rehabilitation protocols for return to play after ACL injury in football players. The purpose of this review is to synthesize information, both classic and new, to aid orthopedic surgeons in treatment of football players with ACL injuries.

Recent findings: Recent studies have demonstrated that not all ACL injuries are alike. Management should be a shared decision-making process between athlete and surgeon. Studies have demonstrated low failure rates when using bone-patellar tendon-bone (BTPB) autograft which is the most common graft utilized for elite football athletes. Outcomes are continually being evaluated by multicenter study groups such as the Multicenter Orthopaedic Outcome Network which has established a thorough

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rehabilitation protocol focusing on athlete milestones. ACL tears in the football athlete are common and challenging injuries. Treatment revolves around ACL reconstruction, most commonly with BTPB autograft. Post-operative rehabilitation is essential and should focus on objective criteria rather than time elapsed. Return to play criteria rely upon symptoms, athlete confidence, strength, and both functional and football specific testing. Return-to-play timelines are individualized for each football athlete but recent trends have highlighted a more delayed return of at least 7-9 months. Rate of returning to play varies from 63-82% and depends upon many factors including level of play and position.

Assessing the validity of wearable inertial sensors in evaluating joint kinetics and hamstring musculotendon mechanics at various running speeds. Medicine and Science in Sports & Exercise, Ahead of Print: 1-50, 2025.

Purpose: Integrating musculoskeletal (MSK) modelling with inertial measurement units (IMUs) offers a promising approach for analyzing joint and muscle function during locomotion. This study examined the validity of combining IMUs, MSK modelling, and inverse dynamics to estimate lower-limb joint moments and hamstring musculotendon (MT) mechanics during treadmill running at varying speeds.

Methods: Eighteen healthy young adults ran on a treadmill at 70% (5.21 ± 0.62 m/s), 80% (5.96 ± 0.71 m/s), 85% (6.33 ± 0.76 m/s), 90% (6.70 ± 0.80 m/s), 95% (7.07 ± 0.84 m/s), and 100% (7.44 ± 0.89 m/s) of their maximal sprinting speed. Kinematic data were simultaneously collected using both an optical motion capture (OMC) system (Vicon) and an IMU system (Xsens), while electromyographic (EMG) data recorded hamstring activity. MSK modelling was applied to both kinematic measurements to calculate lower-limb joint moments and hamstring MT mechanics, with estimated muscle activations validated against the EMG data.

Results: IMU-based estimations closely matched OMC-based calculations, with coefficient of multiple correlations (CMC) exceeding 0.85 for hip and knee joint moments during swing and 0.95 for hamstring MT kinematics across full stride cycles at all speeds. MT force estimations varied among hamstring muscles, with semimembranosus showing the highest agreement (0.96 < CMC < 0.98) across all speeds.

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Linear mixed models showed for each 1 m/s speed increase, root mean square errors between the two systems increased by less than 0.25 N/m for joint moments and 0.05 BW for hamstring MT forces.

Conclusions: IMU-MSK integration is a valid alternative to OMC for estimating sagittal-plane joint moments and hamstring MT mechanics during treadmill running, though differences in peak hip moment during terminal swing warrant caution in field-based applications.

Associations of collegiate football career and incident concussion with players' health: A longitudinal study from the CARE Consortium. Sports Medicine, Ahead of Print: 1-10, 2025.

Background: The influence of repetitive head impacts on collegiate football players remains unclear as prior research is often limited to small samples or short-term studies focused on single seasons.

Objective: Our objective was to determine the associations between collegiate football career or incident concussion and changes in neurocognitive function, postural stability, and physical and psychological health.

Methods: In total, 574 football players enrolled in the Grand Alliance Concussion Assessment, Research and Education (CARE) Consortium (median age 18.0 years [interquartile range 18.0-19.0], 52% white race, 26% with incident concussion) completed baseline and exit evaluations (i.e., beginning and end of collegiate career) consisting of neurocognitive, postural stability, and physical/psychological health assessments, specifically, Immediate Post-Concussion Assessment and Cognitive Testing, Standardized Assessment of Concussion, Balance Error Scoring System, Sport Concussion Assessment Tool-5 (SCAT-5) Symptom checklist, and the Brief Symptom Inventory-18. Adjusted linear regression models incorporating inverse probability of attrition weighting were used to compare changes in scores between baseline and exit evaluations overall and by incident concussion status.

Results: Overall, athletes had small improvements in neurocognitive functioning and postural stability over time but had small increases in symptom severity. Both the incident concussion and no incident concussion groups improved similarly on neurocognitive and postural stability measures (all $p > 0.05$ for difference in change over time between incident concussion groups). Individuals with incident

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concussion reported fewer symptoms and lower symptom severity over time than did those without incident concussion (SCAT symptom count difference - 1.22; 95% confidence interval [CI] - 1.89 to - 0.54; SCAT symptom severity difference: - 2.46; 95% CI - 4.06 to - 0.86; Brief Symptom Inventory-18 somatization difference: - 0.55; 95% CI - 0.93 to - 0.17).

Conclusions: Overall, collegiate football players demonstrated small, non-clinically meaningful improvements in neurocognitive function and postural stability. Moreover, athletes who experienced a concussion reported slight improvements in physical/psychological health symptoms over their collegiate careers.

Brachial plexus injuries in the contact athlete: A narrative review. Annals of Joint, Ahead of Print: 1-11, 2025.

Background and objective: Brachial plexus injuries (BPIs) are a rare but potentially devastating injury among contact athletes. More common injuries, such as "burners" or "stingers", indicate a temporary and less severe insult to either the nerve roots or plexus. The aim of this review is to discuss this spectrum of injuries in their epidemiology, mechanism in contact sports, diagnosis, and treatment.

Methods: This literature review utilized key terms to search manuscripts available in English from inception to October 2024 within three research databases.

Key content and findings: BPIs among contact athletes occur on a spectrum from neuropraxia to neurotmesis. Neuropraxia is very common among contact athletes with approximately a 49-65% incidence among career football players with the most common mechanism being a traction injury. Most of these resolve within minutes to hours. Axonotmesis and neurotmesis can be far more severe injuries and require further investigation if not resolved by 2-3 weeks. The majority of athletes who suffer neuropraxic injuries will return to sport with minimal to no time missed, however, more severe injuries portend variable outcomes.

Conclusions: BPIs are common among contact athletes and involve a spectrum of nerve damage, of which most are self-limiting. However, further evaluation is warranted when symptoms last beyond 2-3

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weeks. Future studies should focus on treatment algorithms and long-term outcomes, including return to sport.

Brain doping substances: Prohibited or not in sports? Biology of Sport, 42(4): 189-202, 2025.

Abstract - There is increasing interest in the area of brain doping among athletes. The term 'brain doping' refers to enhancing mental performance through the non-medical use of specific pharmacological substances, such as drugs and medicines. Despite hundreds of studies, it is challenging to specify well-defined and described substances that could contribute to improving cognitive abilities in sports. Therefore, we systematically reviewed existing scientific literature to identify 'brain doping' substances with proven beneficial effects on sports performance. Only 9 research studies, consisting of 7 interventional and 2 case studies, have investigated the current knowledge on substances that have or may affect the brain at the cognitive level in the context of sports. Simultaneously, we reviewed the Polish dietary supplement market to identify active ingredients promoting cognitive functions and examine their properties related to anti-doping rules. Results from the market analysis revealed 34 substances/extracts of plant/mushroom origin and 120 synthetic substances with potential procognitive properties. Among the synthetic substances, 45 were labelled as 'unclear', including 19 presumed to 'meet WADA criteria'. In conclusion, the term 'brain doping' currently serves more as a marketing tool than as a concept describing a group of substances with scientifically proven cognitive effects. Substances marketed to consumers as 'brain doping' include substances prohibited in sports, as mentioned on the WADA Prohibited List, and those permitted for use by athletes or of unclear status. The open-ended nature of the list of prohibited substances poses challenges in definitively determining the status of many substances found in dietary supplements or medications, which may lead to unintentional violations of anti-doping rules.

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Brief report: Exploring the icing the kicker strategy in the NFL. Journal of Applied Sport Psychology, Ahead of Print: 1-10, 2025.

Abstract - Icing is a strategy used in American football during the last moments of game when a coach calls for a time-out to allow an opposing kicker to supposedly ruminate before a decisive field-kick. Using archival data of pressure kicks from twenty-five National Football League seasons (1999–2024), a generalized estimating equation (GEE) model accounting for nested data was applied. It was found that icing was successful in reducing scoring while other factors such as location (home vs. away), physical environment (indoors vs. outdoors) or season type (regular vs. playoffs) were not associated with the kicks' success. In a secondary analysis it was demonstrated with another limited set of nullified kicks that the second attempt in close time proximity did not yield a significant deterioration in performance thus presently failing to lend support to the "twice taxed" hypothesis (Goldschmied et al., 2010) of why icing works. Theoretical and applied implications are discussed. Lay Summary: Attempts at icing or the calling of time-outs by opposing coaches against kickers in the National Football League before crucial kicks were demonstrated to be successful (1999–2024 seasons). Other factors such as home vs. away, indoors vs. outdoors or regular vs. playoffs were not associated with the kicks' success.

Implications for Practice

- Opposing coach should preserve and use time-outs to ice the kicker before crucial kicks.
- Kickers should practice kicking with icing manipulations in training.

Causes of different force-velocity relationships in sprint running depending on horizontal loads and profiling methods. Medicine and Science in Sports & Exercise, 57(5): 1032-1042, 2025.

Purpose: We aimed to clarify how the horizontal force-velocity (Fvh) relationship during over-ground sprint running differs with horizontal resistance loads and profiling methods (multiple- and single-trial methods).

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Methods: Twelve males performed sprint running (one unresisted and five resisted) using a motorized loading device. During the trials, the ground reaction forces at every step were obtained using a 50-m force plate system. The step-averaged Fvh relationships were then determined using single- and multiple-trial methods with linear and curvilinear models. The differences in Fvh parameters between loading conditions and between profiling methods, as well as the goodness of fit of the regression models to the measured data, were examined.

Results: We found that Fvh plots in each loading condition almost overlapped during acceleration, whereas the horizontal forces deviated toward a lower value around maximal velocity; the linear Fvh parameters derived using the single-trial method had a load dependency; the linear Fvh relationship derived from the multiple-trial method had a bias toward lower force values with less negative slopes compared with the single-trial method; the curvilinear models fitted the pooled data of all loading conditions better than the linear model; and the Fvh relationship within the velocity range of unresisted sprinting was almost linear.

Conclusions: The results of this study indicate that the reported load dependency of Fvh parameters is mainly due to large horizontal forces at very low velocities in resisted sprinting, and the profiling method dependency is mainly due to the attenuation of horizontal force around the maximal velocity of each loading condition. Factors of deviations from a linear Fvh relationship in horizontal force and the validity and usefulness of nonlinear models require further investigation.

Chronic traumatic encephalopathy neuropathologic change is not associated with suicide in former athletes. Scandinavian Journal of Medicine & Science in Sports, 35:e70086, 2025.

Abstract - There are important knowledge gaps and misunderstandings relating to chronic traumatic encephalopathy (CTE). A systematic review and meta-analysis, published in April of 2025, was designed to examine the prevalence, risk factors, and potential outcomes of CTE neuropathologic change (CTE-NC) in former athletes and concluded that suicide was a potential outcome of CTE-NC (Qi et al., Scandinavian Journal of Medicine & Science in Sports. 2025;35(4):e70047). The review included eight eligible postmortem studies, four of which were used to estimate a possible association between the

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postmortem neuropathology and suicide as a manner of death. The authors of two of those four studies did not report a statistically significant association between suicide as the manner of death and having CTE-NC identified in brain tissue. In the largest of the four studies, the brain donors with CTE-NC were statistically significantly less likely to have suicide as their manner of death than donors who did not have CTE-NC. Considering the four studies included in the systematic review, other studies of CTE-NC not included in the review, and the broader literature relating to suicidality in former athletes, the best available evidence suggests that CTE-NC is not associated with depression, suicidality, or suicide.

College-to-NFL stadium turf transitions as a risk factor for lower extremity non-contact injuries in rookie players: A 13-year cohort analysis. Healthcare, 13:1415, 2025.

Background/Objectives: Lower extremity non-contact injuries (LE-NCIs) pose a significant burden on the National Football League (NFL), with ongoing debates regarding playing surface safety. The stressful college-to-professional transition period for rookies, which can include adapting to new playing surfaces, may influence injury susceptibility. This study aimed to determine whether the transition in a home stadium turf type (natural grass, artificial, and hybrid) from the final college season to the rookie NFL season impacts LE-NCI likelihood.

Methods: A retrospective cohort study analyzed 826 first and second-round NFL draft picks from 2012 to 2024. Data on college/NFL home surfaces (defining six transition types), position group, college training surface access, and rookie season LE-NCIs were collected from public sources. Competing risk analysis was used to estimate the cumulative LE-NCI incidence. Multivariable logistic regression assessed the association between turf transition and LE-NCI risk, adjusting for position, draft cohort, and college training access.

Results: During their rookie season, 21.2% (175/826) of players sustained an LE-NCI. Skill position players had significantly higher adjusted odds of LE-NCI compared to hybrid players (AOR = 1.88; 95% CI: 1.20-2.97; p = 0.006). No specific turf transition category showed a statistically significant association with LE-NCI risk compared to the Grass-to-Grass reference in adjusted models. College training surface access was also not significantly associated with risk (AOR = 0.97; 95% CI: 0.65-1.45; p = 0.874).

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Cumulative LE-NCI incidence reached 33.1% by season end, with risk accelerating between weeks 4 and 10.

Conclusions: Home stadium turf-type transition from college to the NFL was not significantly associated with LE-NCI risk in this rookie cohort, suggesting that surface transitions may not be a primary risk factor during the professional transition period. However, our analysis revealed significant position-dependent injury patterns (skill players: AOR = 1.88) and a temporal clustering of injuries between weeks 4 and 10, indicating that rookie LE-NCI prevention strategies should prioritize position-specific interventions and enhanced monitoring during the early- to mid-season high-risk period rather than surface transition-based approaches.

Data everywhere, insight nowhere: A practical quadrant-based model for monitoring training load vs response in elite football. Sport Performance Science Reports, May: 1-14, 2025.

Overview: The paper proposes a practical, quadrant-based framework for monitoring training load and physiological responses in elite football players. It addresses the challenge of overwhelming data in sports science by organizing monitoring tools into a structured model to optimize training, enhance performance, and minimize injury risk. The model emphasizes balancing training load (input) with physiological responses (output) to achieve the minimal effective dose while avoiding diminishing returns.

Conceptual Framework: The Four-Quadrant Model

The framework is built on two key dimensions (see Figure 1):

1. **Training Load vs. Response/Adaptation**:
    - **Training Load**: The physical or physiological stress imposed on the body (input), e.g., volume, intensity, or frequency of training.
    - **Response/Adaptation**: The body's acute response and long-term adaptation (output), e.g., fatigue, recovery, or performance changes.

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2. **Physiological Domains**:
    - **Metabolic**: Related to energy systems, cardiovascular, and aerobic/anaerobic capacities.
    - **Neuromuscular**: Related to muscle function, strength, and movement mechanics.

These dimensions form four quadrants:

- **Metabolic Load**: Measures cardiovascular and energy system stress (e.g., heart rate [HR], VO_2).
- **Neuromuscular Load**: Tracks mechanical and muscular stress (e.g., GPS-based metrics like distance or sprint counts).
- **Metabolic Response/Adaptation**: Assesses cardiovascular and energy system adaptations (e.g., submaximal HR response).
- **Neuromuscular Response/Adaptation**: Monitors muscle recovery and performance changes (e.g., jump tests, subjective wellness ratings).

Key Findings

- **Data Imbalance**: Current practices heavily rely on GPS for load monitoring, which is practical but limited in capturing neuromuscular load. HR monitoring, though valuable for metabolic load, is underused due to practicality issues.
- **Quadrant Strengths and Weaknesses**:
    - **Metabolic Load**: VO_2 is the gold standard but impractical; HR is a viable proxy despite limitations.
    - **Neuromuscular Load**: The weakest quadrant, as GPS metrics often fail to accurately reflect neuromuscular strain.
    - **Responses/Adaptations**: These quadrants are stronger, with multiple tools (e.g., wellness questionnaires, submaximal tests) available to guide training adjustments based on player responses.
- **Recommended Minimum Setup**: A cost-effective, elite-level monitoring system should include:

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- GPS with HR capability for load monitoring.
- At least one variable per quadrant (metabolic load, neuromuscular load, metabolic adaptation, neuromuscular response/adaptation) to ensure comprehensive coverage.

- **Tool Evaluation**: Monitoring tools are categorized by validity, practicality, and cost (color-coded from green [ideal] to red [impractical/limited]). Recommended tools are marked with circled stars in Figure 1.

Practical Implications

- The model helps practitioners prioritize essential metrics, reducing data overload and focusing on actionable insights.
- By monitoring both load and response across metabolic and neuromuscular domains, coaches can tailor training to individual player needs, optimize performance, and reduce overtraining risks.
- The framework supports a “minimum viable monitoring system” for elite football, balancing scientific rigor with practical constraints.

Conclusion: The quadrant-based model provides a structured, practical approach to monitoring elite footballers, addressing the gap between abundant data and actionable insights. It advocates for a balanced, minimalistic monitoring system that covers both metabolic and neuromuscular domains, ensuring effective training load management and player adaptation tracking.

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Dietary carbohydrates influence the performance outcomes of short-term heat acclimation. Medicine and Science in Sports & Exercise, Ahead of Print: 1-40, 2025.

Purpose: Studies on heat acclimation (HA) involving high-intensity exercise report impairments in time trial (TT) performance in the heat immediately after HA. This study aimed to determine whether a high-carbohydrate (HC) diet can enhance 3.22 km run TT performance in the heat following exercise-intensive short-term HA.

Methods: Fourteen healthy military-age males were assigned to either a HC diet (70 %kcal carbohydrate (CHO), n = 7) or lower CHO comparison (COMP, 35 %kcal CHO, n = 7) to be consumed during and for one week after a 6-day HA protocol.

Results: Baseline descriptive statistics, heat tolerance, and run performance were similar between groups ($p > 0.05$). Both groups acclimated, showing reduced heart rate, rectal and skin temperature, and increased sweat rate (each $p \leq 0.03$) during heat stress testing one day and five days after HA. The HC group demonstrated a faster TT (959 ± 103 sec vs. 1067 ± 172, $p = 0.02$, $g = 0.71$) than COMP one day post-HA. HC showed improved TT performance from baseline at one day ($p = 0.01$, $g = 0.59$) and five days post-HA ($p = 0.04$, $g = 0.59$). The HC group showed greater TT improvement at one day (-7.0 ± 4.9 % vs. -0.7 ± 4.7 %, $p = 0.03$, $g = 1.23$) and five days post-HA (-8.4 ± 4.9 % vs. -2.8 ± 3.9 %, $p = 0.01$, $g = 1.18$), with no group differences in HA outcomes or relative strain during TT.

Conclusions: Insufficient CHO intake during heavy daily exertion in unacclimated heat stress, such as in short-term HA, may affect post-HA performance outcomes unless addressed with additional CHO consumption.

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Dilemma in the treatment of sports injuries in athletes: Tendon overuse, muscle strain, and tendon rupture. Scandinavian Journal of Medicine & Science in Sports, 35:e70026, 2025.

Abstract - Injuries to the musculoskeletal system are frequent in elite sports and they are detrimental to athletic performance. This can be due to, for example, (1) overuse disorders of tendon (tendinopathy) that not only lower the training efficiency but also, in many cases, are career-ending for the athlete due to pain; (2) acute muscle strain injury that both causes prolonged absence from competition and results in many re-injuries; or (3) tendon rupture that, apart from a very long rehabilitation period, will also result in many athletes never resuming their original high level of competitive sports. For all three injury examples, several evidence-based prevention programs and treatments are available, and yet the incidence of these injuries remains high and single treatments often provide only partial recovery. In this paper, we highlight the current treatments of these three conditions and focus on the unsolved dilemmas that exist in these sports injuries.

Effect of newly required American football headgear on thermoregulation during exercise in the heat. Sports Engineering, 28:19, 2025.

Abstract - The Guardian Cap (GC) football helmet cover is designed to reduce head impact severity, but the influence on body temperature during exercise heat stress is unknown. We determined whether the GC NXT accentuates hyperthermia in a thermal manikin model and exercising humans (n = 12 males, 18–35 yrs, > 91 kg). The manikin assessed heat loss, heat absorption, thermal resistance, and evaporative resistance in four helmets with and without the GC NXT. The human study recorded rectal (T_r) and skin (T_{sk}) temperature, sweat rate, heart rate (HR), thermal sensation (ThS), rate of perceived exertion (RPE), and fatigue during five 90 min exercise trials in the heat (two helmet models each with and without white and black GC NXT's). In the manikin, heat loss was lower by 2.35 W/m^2 with all helmets when the GC NXT was worn (p = 0.0283). The effect of the GC NXT was small compared to the variation between helmet models (Hedges' g^* = 0.13). T_r, T_{sk}, HR, and sweat rate were similar between trials in humans. Helmet temperatures were 5–10 °C lower with the GC NXT compared to the bare helmet (p < 0.001). ThS was lower with the white (p = 0.007) and black (p = 0.011) GC NXT, but RPE and fatigue were similar.

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Minor differences in heat loss and absorption were found in the manikin, but did not translate into altered thermoregulatory parameters in humans during intense exercise heat stress. The GC NXT did not significantly alter exertional hyperthermia, which can inform decision-making for product use.

Effects of different velocity loss thresholds in full squat with and without blood flow restriction on strength gains, neuromuscular adaptations, and muscle hypertrophy. Scandinavian Journal of Medicine & Science in Sports, 35:e70090, 2025.

Abstract - This study aimed to analyze the effects of four full-squat (SQ) training programs that differed in the blood flow condition [free flow (FF) versus restricted (BFR)] and in the velocity loss (VL) induced within the set (20% vs. 40%) on strength gains and muscle hypertrophy. Fifty-two strength-trained men followed an 8-week (16 sessions) SQ training program from 55% to 70% 1-repetition maximum (1RM) (FF20: n = 14; BFR20: n = 13; FF40: n = 12; BFR40: n = 13). The number of sets n = 13 per session and the inter-set recovery periods (2 min) were matched between groups. A 50% arterial occlusion pressure was applied and maintained during the inter-set recovery for BFR groups. The following tests were carried out before and after the training intervention: (1) cross-sectional area of the vastus lateralis (ACSA); (2) countermovement jump; (3) progressive loading SQ test; and (4) fatigue SQ test. No significant BFR × VL × time interactions were observed. For 1RM and strength-derived outcomes from the progressive loading test, significant VL × time interactions (p = 0.01-0.05) in favor of 20% VL groups were found. Regarding jump performance, a significant VL × time interaction (p = 0.02) also favored the 20% VL groups. A BFR × time interaction (p = 0.02) was observed in favor of the BFR condition for ACSA. Prescribing a certain level of effort through VL results in similar jump and strength performance improvements, regardless of blood flow condition, with optimal gains achieved at a moderate VL threshold (20%). Additionally, the BFR condition maximized muscle hypertrophy compared to FF, making it a valuable strategy for muscle growth.

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Effects of position and injury status on associations between preseason workload and heart-rate variability profiles in American college football players. International Journal of Sports Physiology and Performance, Ahead of Print: 1-8, 2025.

Purpose: To quantify position-specific associations between heart-rate-variability profile characteristics and external-workload parameters in American college football players during preseason camp while accounting for injury-related participation status.

Methods: Twenty-seven players categorized into skill (n = 15) and midskill (n = 12) position groups participated in the study. Players were further categorized as having “full go” (n = 15) or “go as can” (n = 12) participation status. The camp mean and coefficient of variation (CV) (indexing daily variation) for various workload parameters and resting heart rate and the natural log of the root mean square of successive differences (LnRMSSD) were quantified.

Results: Skill players who consistently performed higher total workloads (total distance and player load) exhibited the most stable LnRMSSD (ie, lower CV; $P < .05$), while midskill players with the highest total workloads exhibited greater daily variation in LnRMSSD (ie, higher CV; $P < .05$). These associations persisted after injury-status adjustment. However, consistent between position groups, players with greater daily variation in select workload parameters also showed greater daily variation in LnRMSSD ($P < .05$). No between-groups differences for injury status were observed ($P > .05$), although effect-size analysis showed that LnRMSSD was moderately greater in “full go” players (effect size = 0.73, $P < .07$).

Conclusions: Daily LnRMSSD variation better reflected position-specific workload responses, whereas mean LnRMSSD may affect or be affected by injury status. While skill players tolerated high total workloads with stable LnRMSSD, midskill players exhibited less day-to-day variation in LnRMSSD and high-intensity outputs at moderate workloads.

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Focused review of sleep wearable technology and its application in sports performance. Current Sports Medicine Reports, 24(6): 171-176, 2025.

Abstract - Sleep is a critical component of quality of life, recovery, and performance both physically and cognitively. Tracking sleep with wearables is increasingly prevalent among individuals; the data that are retrieved from these wearables, however, is of unclear relevance. To update this space, literature was reviewed to examine the implications for good and bad sleepers, the current shortcomings of wearables, their current status usage and literature reports in sports medicine, and the future benefits that these devices may have for athletes and nonathletes alike. Wearables have good performance in detecting sleep but have greater difficulty in discerning between quiet wakefulness and early stages of sleep. The use of summative scores in consumer wearables is opaque, which should make sports medicine providers skeptical of their clinical utility. There is ample opportunity for wearables to be clinically useful and research opportunities to better explore the bidirectional relationship between daytime athletic performance and sleep.

From the Field to the Screen: A scoping review of video analysis as a tool for understanding thigh muscle-tendon injuries in football. Clinical Journal of Sport Medicine, Ahead of Print: 1-12, 2025.

Objective: Muscle-tendon injuries are critical setbacks in professional football, significantly affecting player availability and team performance. Understanding these injury mechanisms through video analysis is crucial for developing effective prevention and rehabilitation strategies that enhance player welfare and optimize performance. This review aimed to synthesize data from articles that used video analysis to explore mechanisms of thigh muscle-tendon injuries in football.

Data source: A comprehensive literature search was conducted from 2010 to 2025 using a scoping review methodology. The quality of the included studies was assessed using the Quality Appraisal for Sports Injury Video Analysis Studies (QA-SIVAS) scale.

Main results: The review of 10 studies identified that noncontact and indirect contact mechanisms are predominantly responsible for severe thigh muscle-tendon injuries in football. Common injury scenarios

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involved sprinting-induced strains and kicking actions, highlighting the significant role of eccentric loading and rapid biomechanical changes.

Conclusions: Video analysis has emerged as a vital tool in sports medicine, providing deep insights into the complex mechanisms of thigh muscle-tendon injuries in football. The continuous improvement of analytical methods, including the adoption of advanced technologies such as artificial intelligence, is imperative for refining prevention and rehabilitation protocols.

Improving sleep quality to enhance athletic activity – The role of nutrition and supplementation: A mini-short review. Nutrients, 17:1779, 2025.

Abstract - Sleep is a fundamental part of life for all living beings. The propensity to fall asleep is regulated by a circadian rhythm, which controls the secretion of the hormone melatonin. Melatonin secretion is linked to the light and dark phases of the day/night cycle. Getting proper sleep is an essential part of a healthy lifestyle. Scientific evidence shows that sleeping less than 7 h per night, or as little as 2 h per night, is a cardiovascular, metabolic, and cerebral risk factor. In addition, the role of sleep is fundamental for the recovery phase for athletes. Nutrition, supplementation, and exercise can greatly support the quality and quantity of sleep. They can have positive effects on sleep through specific physiological and biochemical responses. The objective of this short review is to critically investigate the possible sleep benefits of nutrition, exercise, and supplementation and to discuss further directions for research in this area.

Irregular sleep/wake patterns in student-athletes exposed to early morning training. Journal of Sports Sciences, Ahead of Print: 1-11, 2025.

Abstract - This study aimed to examine the sleep parameters and sleep/wake regularity of a cohort of student athletes who start training between 06:30 and 07:00. Twenty-one male Rugby Union players, aged 21 ± 2 years and competing at a national level, were assessed using actigraphy over two weeks, and the Athlete Sleep Screening Questionnaire (ASSQ). Sleep/wake regularity was calculated using the

Sleep Regularity Index (SRI). Wilcoxon signed-rank tests showed that nocturnal sleep preceding morning training had a significantly shorter sleep duration (1.8 hr, r = .67), and advanced sleep onset (0.9 hr, r = .50) and sleep offset times (3.2 hr, r = .85) compared to nocturnal sleep preceding free days. The variability of training demands resulted in an inconsistent sleep pattern between consecutive days, resulting in a median SRI score of 67.0 (interquartile range: 17.0). Pearson correlations revealed that lower SRI was significantly associated with a higher daily sleep duration including naps (r = -.62), delayed sleep onset (r = -.50) and sleep offset (r = -.60), and a later chronotype assessed using the ASSQ (r = .52). These findings indicate that early morning training is a factor contributing to irregular sleep/wake patterns in student-athletes, and where feasible should be scheduled at an alternative time.

Kinematics and kinetics of pick-up acceleration. International Journal of Sports Science & Coaching, Ahead of Print: 1-9, 2025.

Abstract - In team sports, linear acceleration is a fundamental skill defining success at a competitive level. While maximal acceleration capability from a static start has been studied extensively, team sport athletes often accelerate from a submaximal speed. This ability to maximally accelerate from a non-static start, in coaching terms “pickup acceleration,” is largely unexplored. In this narrative review, pickup acceleration is defined, and a framework for step analysis is presented. The kinematics, kinetics, and technical demands of pickup acceleration are discussed, and normative data around the pickup steps are highlighted. The practical applications of the findings are then discussed, and future research directions are outlined.

Maximum-speed single-leg bridge test for concentric functional assessment and exercise in an athlete with recurrent hamstring strain injuries: A case report. International Journal of Sports Physical Therapy, 20(5): 716-726, 2025.

Introduction: The high recurrence rate of hamstring strain injuries (HSI) suggests that using only muscle strength and flexibility differences between the injured and uninjured sides may be insufficient. Recurrences are frequent during high-speed movements; however, no assessment method currently

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exists to evaluate high-speed functional performance. This case report aimed to highlight the use of the Maximum-Speed Single-Leg Bridge Test (MS-SLBT) in a rugby player with a fourth recurrence of HSI to identify functional impairments that are not detectable with conventional assessments, and to demonstrate that these impairments can be improved by utilizing the test as an intervention.

Case description: A 21-year-old university rugby player with a history of three prior HSI experienced a fourth recurrence in the semitendinosus. Magnetic resonance imaging (MRI) confirmed a Grade I injury, and ultrasound images showed scarring and tenderness of the muscle fibers. At one month post-injury, the subject reported no pain during hamstring stretching, isometric or eccentric contractions, or sprinting. The subject was unable to complete the MS-SLBT because of severe pain and fear. After two months, the pain during the test decreased, but the buttock-raising speed and height remained limited. The subject then performed the MS-SLBT exercise as a daily intervention for two months and the effects of training and detraining were evaluated.

Outcome: After the intervention, there was a 76% improvement in buttock-raising speed, 84% improvement in height, 34% increase in isometric knee flexor torque, and 97% increase in flexibility. These improvements were maintained even after a 1.5 month of detraining period. Ultrasound images showed reduced muscle fiber irregularities, improved centralization of muscle contractions, and increased contraction speed.

Discussion: The MS-SLBT is effective in identifying residual functional impairments in athletes who meet conventional return-to-play criteria after HSI. Furthermore, MS-SLBT may serve as an effective intervention to improve these functional deficits within one month.

Monitoring dietary supplements for undeclared diuretics as a strategy for preventing unintentional doping. Processes, 13:79, 2025.

Abstract - Diuretics are essential medicines used in the management of hypertension, heart failure, edema, and other medical conditions. Due to their effectiveness in promoting water loss, many cases of undeclared diuretics in dietary supplements (DSs) have been reported. Most of these DSs were marketed for “weight loss promotion” and “detoxification”. Diuretics are regarded as doping

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compounds as well because of their significant potential to cause weight loss or mask the usage of other doping compounds. The intake of DSs that contain undeclared diuretics can result in serious consequences, including side effects, drug–drug interactions, and the violation of anti-doping rules if such products are taken by professional athletes. The aim of the present study was to develop a rapid, sensitive, and cost-effective high-performance liquid chromatography (HPLC) method for monitoring DSs for undeclared diuretics. The validated method allows for simultaneous monitoring for furosemide, hydrochlorothiazide, and torasemide. The LOD and LOQ were 0.78 µg/mL and 2.38 µg/mL for hydrochlorothiazide, 0.99 µg/mL and 3.01 µg/mL for torasemide, and 0.69 µg/mL and 2.08 µg/mL for furosemide. The method was successfully utilized to screen ten DSs for undeclared diuretics. Contamination was detected in only one sample. These results underscore the critical need for regulatory oversight and independent quality control of DSs. Implementing routine screening could significantly reduce the risk of adverse health effects and unintentional doping among athletes. Ultimately, this study contributes to enhancing consumer safety and ensuring fair competition in sports.

No association between fencing response after concussion and recovery time, concussion severity, or clinical outcomes in professional American football players. Neurosurgery, Ahead of Print: 1-8, 2025.

Background and objectives: The relationship between tonic posturing (also known as the "fencing response") after concussion and clinical outcomes is unclear. The objective of this study was to compare clinical outcomes between professional American football players with a documented fencing response after concussion to players without documented fencing response after concussion.

Methods: Players with a fencing response after concussion were matched 1:2 to players without documented fencing response after in-game concussion by age, position, and concussion history from the 2018 to 2023 seasons. The primary outcome measure was number of days until return to full participation. Secondary outcome measures included screening of neurocognitive function with the Standardized Assessment of Concussion (SAC) and modified Balance Error Scoring System (mBESS), and subjective symptom reporting with the Sport Concussion Assessment Tool-5. An independent-samples median test compared game-day SAC and mBESS outcomes between groups. A Kaplan-Meier curve

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evaluated fencing response after concussion on days until return to full participation. A linear mixed-effects model for 11 Sport Concussion Assessment Tool-5 total symptom scores (ie, day of injury up to postinjury day 10) compared symptom recovery over time between fencing groups.

Results: Fifty-seven fencing responses were documented over 6 seasons. Nearly 2-in-3 players with a fencing response played defensive back or wide receiver. There was no difference between players with a fencing response (median = 10 days) and those without fencing response (median = 9 days) in days until return to full participation (P = .16). Fencing response was not associated with a change in hazard for days until return to full participation ($\chi 2$ = 0.91; P = .34). There was no difference between groups in SAC scores (P = .18-.99) or mBESS score (fencing: 4, nonfencing: 4; P = .87). There was no difference between groups in symptom scores over time (β: -0.02; P = .92).

Conclusion: Fencing response after concussion was not associated with symptom recovery, concussion severity, or short-term recovery in professional American football.

Return to performance: Machine learning insights into how absence time following muscle injuries affects match running performance in LaLiga soccer players. Biology of Sport, 42(4): 275-286, 2025.

Abstract - To determine how absence time after muscle injuries affects external load metrics in elite soccer players and identify which performance variables are most impacted by the injury. A total of 110 lower limb muscle injuries from LaLiga players were analyzed. Following an analysis of pre- and post-injury data to identify which outcomes were affected by muscle injury, machine learning algorithms were employed to examine relationships between absence duration and performance metrics. Maximal speed, maximal acceleration, maximal deceleration, composite index (i.e., overall player performance) and sprint count during matches were the most affected variables after return to play. The multiple linear regression (MLR) model and random forest regression (RFR) presented an R^2 of 0.348 and 0.442. Maximal speed was the variable most strongly associated with absence time in both models (coefficient in MLR = 7.94; mean absolute SHAP value in RFR model = 4.99), with longer recovery periods correlating with reduced match performance in this metric. Maximal acceleration and deceleration also showed declines with increased absence time. In contrast, sprint count exhibited no significant relationship with

absence time. Maximal speed, acceleration and deceleration capacity, as well as sprint count and overall performance, are affected after muscle injuries. However, prolonged recovery following muscle injuries especially reduces maximum speed and acceleration/deceleration capacity in elite players during matches, while sprinting actions remain unaffected by absence time.

Squatting biomechanics after femoroacetabular impingement surgery: A systematic review and meta-analysis. Clinical Journal of Sport Medicine, Ahead of Print: 1-12, 2025.

Objective: Femoroacetabular impingement syndrome (FAIS) is a prevalent cause of hip pain, characterized by distinctive motion patterns. However, the impact of surgical intervention on squatting biomechanics remains underexplored. This review aims to evaluate the effects of surgical treatment on biomechanical outcomes during squatting in patients with FAIS.

Data sources: A comprehensive search was conducted across multiple databases following PRISMA guidelines (Prospero registration number CRD42023473974). Data of longitudinal, prospective/retrospective cohorts, case-control, and clinical trial studies with 3-dimensional motion analysis describing kinematics and/or kinetics for any joint during squatting in patients with FAIS comparing presurgical to postsurgical biomechanical data on the same patients were included. Quality assessment was performed using MINORS criteria. Both qualitative and quantitative analyses were performed.

Main results: Six studies met the inclusion criteria for qualitative review and 5 were suitable for meta-analysis. The overall quality of the studies was moderate based on MINORS criteria. The postsurgical motion analysis was performed between 6 and 32 months postoperatively. Although most studies reported no significant changes in kinematic or kinetic variables postsurgery, individual studies reported specific increases in squat depth, pelvic sagittal range of motion, anterior pelvic tilt, hip and knee flexion, and ankle dorsiflexion angles. Meta-analysis revealed no significant differences in the kinematic parameters analyzed.

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Conclusions: These findings underscore the multifaceted nature of FAIS, suggesting that bone deformities alone do not account for the biomechanical limitations observed during squatting in these patients. This highlights the need for further biomechanical investigations.

The effect of heat acclimation on critical environmental limits and rate of rectal temperature change. Journal of Applied Physiology, 138: 1150-1160, 2025.

Abstract - Quantifying the effect of heat acclimation (HA) on critical wet-bulb globe temperature (WBGTcrit) and rate of rectal temperature change (vTre) is relevant for developing guidelines with regards to occupational safety while working in warm environments. This study quantified the effect of HA and the period following cessation of the HA protocol on WBGTcrit and vTre. Twenty-eight non-acclimatized participants were divided into a HA (n = 15) and control (CON; n = 13) group. The HA group underwent a warm-humid (35°C, 65% relative humidity) controlled hyperthermia HA protocol (5-9 days of achieving Tre ~38.5°C for 60 min) and four progressive heat stress tests (HSTs) to identify WBGTcrit and examine vTre: pre-, after 5 and 9 days of HA, and 4 to 8 days of no heat exposure following HA. CON performed two HSTs on average 13 days apart without heat exposure in between. HA increased WBGTcrit after nine (28.5 ± 2.7°C vs. 30.5 ± 2.0°C; P = 0.016) but not 5 days (28.5 ± 2.4; P > 0.05). No effect of HA on vTre was observed (P > 0.05). Four-to-eight days post-HA, WBGTcrit and vTre did not differ compared with 9 days of HA (P > 0.05). However, a reduction in vTre (-0.4 ± 0.3 °C/h) was observed when comparing 4 to 8 days post-HA to pre-HA. In conclusion, our results demonstrate that more than 5 days of HA are required to increase WBGTcrit and indicate that 9 days of HA proceeded by adequate recovery reduced vTre during exercise in the heat.

New & Noteworthy - We assessed the effect of heat acclimation (HA) on critical environmental limits and rate of rectal temperature change. We show that more than 5 days of heat acclimation are required to increase critical environmental limits and that 9 days of HA proceeded by adequate recovery reduces the rate of rectal temperature change. These findings enhance our understanding of heat acclimation's effect on work capacity in the heat and may be used to design occupational guidelines.

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The relationship of infrared thermography with the identification of changes in biological markers after exercise: A systematic review. Journal of Bodywork and Movement Therapies, Ahead of Print: 1-28, 2025.

Abstract - Indirect biological markers of muscle damage are used to measure the level of muscle damage caused by exercise in athletes who are constantly exposed to overloads. However, infrared thermography is an evaluation and diagnostic method that has been frequently used to identify the location of possible injuries caused by temperature asymmetries. A systematic review was carried out with the aim of verifying whether changes in skin temperature can be compared with the inflammatory physiological changes observed after exercise protocols in trained individuals. The PRISMA recommendations were followed in January 2023, in the Pubmed, Medline, Lilacs, Scielo, Sportdiscus and Embase databases. A total of 165 studies were found. After screening, five studies were selected for this systematic review. The results suggest that infrared thermography is being widely used for injury screening, and that few studies associate the method with biomarker tests. However, it may be more advantageous for identifying muscle damage, as it does not determine the location of the injury, but only what is injured. Therefore, the use of infrared thermography as a tool combined with clinical examination techniques may be a complement to clinical evaluations and support decisions.

Training alters joint power distributions during running in National Football League draft preparation players. Sports Biomechanics, Ahead of Print: 1-19, 2025.

Abstract - The aim of this study was to compare joint-specific contributions to total lower limb joint power during a speed run in American style football players before and after a 6-week National Football League (NFL) draft preparation camp. Fifty-two players performed one 5-second treadmill run at 5.5-6.5 m/s before and after camp. Speed was consistent between tests. Bilateral hip, knee, and ankle power generation (positive) and absorption (negative) were calculated from kinematics and kinetics using inverse dynamics. Negative total limb power decreased for both limbs, significantly for the left ($Z = -2.113$, $p = .035$, $r = .30$) due to significant decreases in negative ankle power bilaterally (left, $Z = -1.967$, $p = .049$, $r = .27$; right, $Z = -2.623$, $p = .009$, $r = .36$). The contribution of negative left hip power significantly

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increased (Z = -2.398, p = .016, r = .31). While no other significant changes in positive or negative joint-specific magnitudes or contributions occurred, there were general distal-to-proximal shifts in joint power contributions. Players appear to have adopted more energy efficient running through moderate joint-specific mechanistic training adaptations such as a reduction in ankle power absorption.

Under pressure: Integrating machine learning to quantify clutch moments in American football. International Journal of Sports Science & Coaching, Ahead of Print: 1-9, 2025.

Abstract - This study introduces a novel pressure index framework designed to quantify the intensity of crucial moments in an American football game. Our model integrates game-specific metrics such as time remaining, score differential, field position, and play context to establish quantitative measures of pressure on offensive and defensive units. The two proposed indices, viz, the composite offensive pressure index (PIO3) and defensive pressure index (PID) dynamically capture the evolving situational stress during a game. To validate and assess their predictive power, we employ machine learning techniques, including Logistic Regression, Random Forest, and Gradient Boosting models. Using play-by-play data from the 2023 NFL season, the results demonstrate that the ensemble learning methods outperform logistic regression, with Gradient Boosting achieving the highest predictive precision (AUC=0.944) in forecasting drive success. This machine learning-driven framework provides new insights into situational dynamics and performance under pressure such as identifying pivotal clutch moments and quantifying critical drive transitions offering potential applications in advanced performance analysis, coaching strategies, and sports analytics research.

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